CHAPTER 1

The Problem and Its Background

Introduction

The Philippines naturally provides the people with abundant inland water resources. This is due to the large body of water that surrounds the land and the multiple islands that the country has. There are 221 lakes and 421 rivers in the country. Moreover, its tropical climate with annual rainfall produces approximately 146 billion cubic meters of freshwater each year, sustaining the country's irrigation system and other municipal and domestic use (Senate Economic Planning Office, 2023). In fact, 64 out of 79 provinces in the Philippines reside in coastal lines (Tuddao & Gonzales, 2024).

However, despite the country's archipelagic nature, presence of water sources does not necessarily provide water that is consistently suitable for domestic use. The increase in demand for freshwater resources driven by growing population and economic demands has reduced availability in supply due to the degradation of quality and inefficient use of resources (Sanchez et al, 2024). Local communities such as Barangay Putingkahoy in Rosario, Batangas, which has a population of approximately 70 families based on the 2020 census, continue to experience challenges related to safe and consistent water access (Francisco et al, 2024). These challenges may lead households to seek for alternative sources such

as artesian wells, traditional hand pumps and rainwater harvests, which are often untreated and unmonitored.

Contaminated water can compromise hygiene practices, increase the spread of waterborne diseases, and negatively affect overall public health and quality of life as water may pose contamination depending on its different sources. Water contaminants are substances that make water unsafe or unsuitable for human use, and they usually originate from both natural and human activities. These contaminants can include heavy metals such as lead and mercury, harmful microorganisms such as *Escherichia coli*, pesticides, and industrial or domestic waste (World Health Organization, 2022). The impact of contaminated water extends beyond consumption, affecting household sanitation, personal hygiene, food preparation, and community health. The Philippine Statistics Authority (2022) reported an annual average of 50,058 water-borne disease cases, including typhoid fever, cholera, and acute diarrheal infections. This highlights that unsafe water remains a persistent concern for public health, particularly in rural and low-income communities.

Globally, the reliance on hand pumps remains a practice for addressing water shortage in low-income countries. An estimated 200 million people in Sub-Saharan Africa depend on about 700,000 hand pump units as their primary source of drinking water (Danert, 2022). In the Philippines, according to the National Water Resources Board (NWRB), almost 11 million families rely on water acquired from hand-pumps outside Metro Manila (Velasco, 2023).

Hand-pumps, locally termed *poso*, play a significant role in providing a reliable and accessible supply of water for drinking, cooking, sanitation, and irrigation. However, this is only possible when the hand pump systems are properly maintained, safe, and functional, as their failure or contamination poses significant risks to water security and public health. Hand pumps are not immune to contamination. Over time, they can act as reservoirs for microbial growth and chemical corrosion, especially when regular maintenance and sanitation practices are neglected (Afonso et al., 2025).

Contaminants such as lead and mercury, harmful microorganisms like *Escherichia coli,* pesticides, and industrial waste make water unsafe and unfit for consumption, often originating from agricultural runoff and improper waste management (Zhang et al., 2023). When these pollutants are present, the contaminated water itself becomes a carrier of pathogens and toxic substances that can cause severe health problems such as gastrointestinal infections, neurological disorders, and long-term chronic illnesses.

A study from Los Baños, Laguna, revealed the presence of microorganisms in water collected samples, with 10 of the households recorded to source *Escherichia coli* from their water pumps (Panganiban et al, 2023). Moreover, a local news report from Philstar (Unson, 2024) reported three (3) deaths and twenty-one (21) residents hospitalized due to a diarrhea outbreak in a remote barangay in Talayan, Maguindanao del Sur. Investigations confirmed that the source of drinking water for the residents was from *posos* sourced from the river

banks, making the residents susceptible to waterborne bacteria. These cases demonstrate that *posos* can pose significant public health risks in the country, especially in communities without access to commercially-treated water.

Studies have also shown that microbial contamination from hand-pumps is higher in contamination exposed to localized pathways than general environmental risks and general upstream sources. Localized pathways refer to the points or routes wherein the contaminants may enter or come into contact with water. In a study conducted by Charles et al. (2025), it was found that 50% of the water samples tested positive for *Escherichia coli* prior to cleaning. Moreover, the water tested negative for the bacteria after cleaning the tap or spout, proving that spouts or taps may be a microbiological reservoir and may serve as a source of pathological diseases after contact with water.

Contamination in water may occur due to various natural and human-related sources. Microbial biofilms can develop inside the internal components of hand pumps, leading to the persistence and reintroduction of pathogens such as *Escherichia coli* and *Salmonella* into the water even after it has been treated (Bhatti et al., 2023). The corrosion of metallic pump parts not only reduces water quality but also contributes to the presence of harmful ions in groundwater, further worsening contamination problems.

Although previous studies have documented that hand pump or *poso* spouts may harbor microbial organisms and pose health risk to communities, there is still

limited information that proves the microbial quality of the spouts in the selected barangays in Bulacan. Existing studies have primarily focused on the water quality after it has been emitted from the *posos* and it has also been recognized as a possible cause of contamination when tested for the quality of water before and after cleaning. However, the direct microbial composition of *poso* spouts and their role in contributing to contamination in communities with no access to water treatment systems remains unclear. Bacterial characterization and profiling of bacterial isolates remains a crucial step in diagnosis of microbial contamination. This analysis enables further assessment in the bacterial population as well as provide reliable data to guide correct environmental interventions (Kgoale et al., 2023).

Therefore, this study aims to isolate and profile bacteria from *poso* spouts in Bulacan, in order to assess the microbial quality and potential risks it creates to the rural community. This study also aligns with the United Nations Sustainable Development Goals (SDGs), particularly Goal 6 (Clean Water and Sanitation), Goal 12 (Responsible Consumption and Production), and Goal 3 (Good Health and Well-being), by providing knowledge on the impact of spouts in affecting the microbiological quality of water and their impact to the community.

Statement of the Problem

Poso spouts or hand-pumped wells remain an important source of water for household use in several communities across the Philippines. However, these water sources are frequently constructed without sufficient protection and are often exposed to environmental contaminants, indicating potential health risks (Bekoe et al., 2025).

The absence of localized bacterial profiling limits the ability of health professionals and local authorities to properly assess the microbial safety of *poso* spout water. In addition, bacterial populations may differ among poso spouts depending on environmental conditions, sanitation practices, and surrounding land use. This variation highlights the need for site-specific investigation (Bekoe, 2025).

Therefore, this study aims to isolate and profile bacterial microorganisms from selected *poso* spouts in Bulacan using standard microbial procedures. Specifically, the study seeks to answer the following questions:

1. What bacterial organisms can be isolated from swabbed samples collected from selected *poso* spouts in Bulacan?
2. What are the cultural, morphological, and biochemical characteristics of the isolated bacterial organisms?
3. Are bacterial indicator organisms and potential opportunistic or pathogenic bacteria present in the collected samples from *poso* spouts?

4. Are there differences in the bacterial profiles among the selected *poso* spout sampling sites?

Objectives of the Study

1. To isolate bacterial microorganisms from swabbed samples collected from selected *poso* spouts in Bulacan.
2. To determine the cultural and morphological characteristics of the isolated bacterial microorganisms through colony morphology and Gram staining.
3. To identify the biochemical characteristics of the isolated bacteria using standard biochemical tests.
4. To compare the bacterial profiles among different selected *poso* spout sampling sites.

Hypothesis

This study will use a 0.05 level of significance ($\alpha = 0.05$) to determine whether to accept or reject the null hypothesis.

H_0: There are no bacterial microorganisms present for isolation and profiling from selected *poso* spouts in Bulacan.

H_1: There are no bacterial microorganisms present for isolation and profiling from selected *poso* spouts in Bulacan.

Significance of the Study

The isolation and bacterial profiling of microorganisms from *poso* spouts is pertinent in the understanding of the state of microbial contamination in community water sources and its implication on public health. The strategy improves rural water safety by evaluating and characterizing the identity of bacteria present in commonly used *poso* systems, particularly in areas with circumscribed access to centralized water treatment, and thereby fostering the generation of novel ideas in treating the water. Moreover, the study is particularly significant as it fixates on the isolation and bacterial profiling of microorganisms present from *poso* spouts in the selected barangays in Bulacan, elucidating potential health hazards associated with untreated water sources and contributing insights on probable strategies for improved water quality management.

For the Community. Particularly the residents of the selected barangays in Bulacan, will directly benefit from the study as it will provide information and awareness about the microbial population of the water obtained from the *poso* spouts, with which they use on a daily basis for routine tasks such as drinking, cooking, and hygiene. The study sheds visibility into the possible health hazards and promotes safe water practices to guarantee the well-being and security of residents by verifying and establishing the identity of the bacteria present in the chosen water sources.

For Local Government Units and Health Organizations. The findings of this study will help guide the LGUs and existing health organizations on possible community-based sanitation initiatives and also in developing programs that tackle water quality.

For Students and Researchers. The study will serve as a useful guide and reference material for future investigation about waterborne pathogens, antimicrobial agents, and water sanitation systems or methods. In furtherance of improving students' practical skills and expertise in microbial isolation and identification, the study will provide a foundation for further research on environmental wellness, water microbiology, and microbial contamination on commonly used *poso* water sources.

For Government and Policy Makers. This evidence-based study can aid in the development and implementation of sustainable and community-adaptable strategies for improving water quality led by municipal, provincial, and even national agencies. This will help in ensuring safe access to clean water across the country for different communities.

For the Public Health Sector. The study endorses efforts towards preventing disease and helps identify possible sources of waterborne maladies. Designing tailored measures to minimize microbial exposure and enhance general public health surveillance can be far easier and feasible with an understanding of the bacterial makeup of *poso* spout water.

For Medical Technologists. The study is of great relevance to medical technologists as it integrates the microbiological principles in the identification and profiling of bacteria from ambient samples. It unifies the roles and responsibilities of medical technologists in the public health scheme, especially in the fields of water analysis, microbe detection and identification, and the prevention of water-related illnesses.

Scope and Limitation of the Study

This study focuses on the isolation and bacterial profiling of microorganisms from selected *poso* spouts in Bulacan. The goal of this study is to determine the type of bacteria present in the spouts of the water pumps. This study involves environmental sampling only and is limited to the swab collection from the *poso* spouts. It will not include any human participation nor require any personal information to be collected from them. A number of 6 *poso* spouts will be investigated by collection of swabbed samples from the spouts directly. The study will be limited to the isolation of bacterial microorganisms by using standard microbial techniques. It is then characterized based on their morphologic characteristics which includes the morphology of the colony and also with the use of gram staining. For presumptive identification of the samples, it will be further subjected to subculturing on enriched and selective media and biochemical tests.

After the isolation and identification of bacterial microorganisms, the bacterial profiles among the selected sampling sites will be compared to look for

similarities and differences when it comes to the presence of bacteria across different locations of the water pumps. The comparison is conducted and analyzed descriptively. The study will only be limited to the selected number of *poso* spouts mainly due to the accessibility, time, and resource constraints. Therefore, the findings of this investigation will not be fully observable in all *poso* spouts in Bulacan. The samples are collected at a chosen single time point, which means that the approach of the study does not take into account the possible changes in bacterial presence over time or seasonal variations.

Furthermore, the study will only focus on culture-based and biochemical identification methods, which means that bacteria that require non-cultural bacteria may not be detected. The researchers will also not utilize molecular techniques like DNA sequencing and polymerase chain reaction. It will only be focused on the isolation and profiling of bacteria while the investigation for antibiotic resistance will not be performed. Furthermore, the study will not evaluate the physical and chemical properties of the water samples from the water pumps. The study will provide baseline information on the bacterial composition of selected *poso* spouts in Bulacan, providing a basis for subsequent and more comprehensive studies in the future.

CHAPTER 2

Review of Related Literature and Studies

Community Hygiene

According to the World Health Organization (2023), approximately 1.7 billion people worldwide depend on potable water sources that are compromised with fecal matter, which increases their susceptibility to waterborne diseases such as cholera, typhoid fever, and diarrhea. Inadequate sanitation and hygiene infrastructure, or the dearth of optimal waste disposal systems, is profoundly associated with the diffusion of waterborne diseases and environmental degradation worldwide (World Health Organization, 2024). This persisting economic stress is further aggravated by substandard means of water management, increasing demographics, poor sanitation practices, and socio-economic status (Alfonso et al., 2022). In the Philippine context, deficient wastewater treatment significantly contributes to the proliferation of waterborne diseases, disproportionately affecting destitute populations (Corpuz, 2025).

The key function of community hygiene and sanitation in guaranteeing and sustaining public health is stressed by the steady subsistence of microbial contamination in rural water sources, such as *poso*. Hence, accentuation on a narrow, specific, and community-level scope of hygiene and sanitation practices are pertinent in effective water pollution management. In multiple barangays, the vulnerability to waterborne diseases is heightened due to the reintroduction of microbial contaminants to the available sources, which is exacerbated by

inadequate sanitation practices and the improper management of water (PIDS, 2024). As stated in the study conducted by Molina et al. (2021) entitled *"Water, Sanitation, and Hygiene Practices in the Philippines: Meeting National and Global Targets at the Local Level,"* households that lack access to basic sanitation amenities like toilets, soapy handwashing stations, and appropriate waste disposal systems are much more likely to experience contaminated water and associated health problems as it can cause microorganisms to reintroduce into formerly clean water systems. Furthermore, the study "*The Last-Mile Challenge: Water, Sanitation, and Hygiene (WASH) in the Philippines"* recognizes the persistent difficulties of acquiring equitable WASH services, emphasizing that the availability of water infrastructure alone does not guarantee adequate sanitation and hygiene, which remain significant bottlenecks in rural and disadvantaged communities (Ulep et al., 2024).

According to Charles et. al (2025), various cases has been recorded that *Escherichia coli* was detected at the point of collection (POC), hence, the contamination did not originate from the water source per se, rather from pathways in regards to the hygiene practices of the locals – such as human or animals' interaction with the spout before the sampling of the water. With that, it suggests that the transmission of the microorganisms was due to the contact made by humans or animals with the tap outlet which, as a consequence, was carried over to the water sample itself.

On the other hand, another sampling method was conducted wherein water that was free from *E. coli* at the point of collection (POC), contamination occurred at the household point of use (POU), such as in cups or storage containers. This shows that despite satisfying water safety standards, improper handling and hygiene practices at the household level has the capacity to negate water quality and ultimately, remain a hazard to the health of the local population. With that, consistency of results were observed despite differences in sampling methods, suggesting that human and animal contact plays a significant role in compromising the integrity of water from the *poso*.

Thus, in rural communities that primarily depend on *poso* water systems or shallow wells, efforts to improve sanitation extend beyond infrastructure development. They further require community engagement and education on hygiene behaviors such as safe water handling and proper handwashing. These approaches discourage the subsequent contamination of water sources after collection and restrict the microbial load that can endanger health. In light of this, the study emphasizes the significance of screening and tracking the status of the microbial communities in the poso water systems utilized by the neighborhood in order to safeguard the integrity of water and improve public health.

Poso-Acquired Infections

In rural regions, hand pumps are a staple for community water access; however, they frequently produce water contaminated with microbes, even though they are officially categorized as "improved" water sources (Kelly et al., 2021).

Studies show that many “improved” water points like tubewells and boreholes are frequently contaminated with fecal indicator bacteria, putting those who use them at risk for waterborne diseases (Kelly et al., 2021). According to Barmase (2025), contaminated hand pump water has been found to harbor specific pathogens such as *E. coli*, *Salmonella*, *Shigella*, *V. cholerae*, *S. aureus*, and *P. aeruginosa*. These organisms are linked to serious gastrointestinal illnesses and represent a major threat to public health. Infants, elderly and immunocompromised individuals who are included among the vulnerable populations, face higher risks of adverse health outcomes from exposure to these pathogens (Thaniem et al., 2022). The transmission of these pathogens is frequently worsened by poor sanitation practices and subpar hygiene, which drive the fecal-oral cycle and allow contaminants to enter water supplies as early as the collection point (Ismail et al., 2024) (Madamanchi, 2024).

Structural flaws in hand pump systems, including degraded seals or rusted piping–further jeopardize water quality by creating pathways for environmental pollutants to infiltrate the supply (Kanyagui et al., 2023) (Kelly et al., 2021). To give an example, environmental events like heavy rain and melting snow can sweep bacteria into well infrastructure; meanwhile, underground leaks from storage tanks or drainage from septic systems can discharge microbes directly into the water supply (Thaniem et al., 2022). According to Thaniem et al (2022), groundwater used for everyday purposes can harbor human-specific pathogens, including *Shigella, Norovirus, Hepatitis A, E. coli, Salmonella, and*

Cryptosporidium; these organisms mainly infect and multiply within the host's digestive system.

Furthermore, private well systems have been found to contain opportunistic pathogens like *Legionella pneumophila* and *Aeromonas hydrophila*. These specific risks are particularly concerning because they are often missed by standard water quality testing protocols (Mimura et al., 2023). The survival of these opportunistic pathogens is frequently attributed to the development of biofilms inside pump parts and delivery pipes. These microbial layers act as a persistent source, releasing bacteria into the water even when the primary source water looks clean (Odeyemi et al., 2023).

Mbachu et al. (2022) suggest that hand pumps are a safer alternative to open-well retrieval systems, as they notably reduce the risk of introducing surface-level contaminants into the water. However, the internal mechanical parts can unfortunately act as a breeding ground for microbes even though hand pumps are designed to be protective. Because of this, it cancels out the safety advantages of having a sealed extraction system (Mimura et al., 2023).

Rural Water Systems

According to *Traditional water management practices in the Philippines* (2025), water management in the Philippines is a shared responsibility among the community, collective decision-making, and must have a strong influence from the community. Managing water as a communal resource is said to be deeply rooted in social relationships, cultural values and laws that revolve among the community

members. In addition to this, Sanchez et al. (2024) stated that many households rely on freshwater resources in the Philippines such as rivers, wells, and hand pumps, which can be exposed to pathogenic microorganisms or contaminated fecal matter. Sanchez et al. (2023) stated that poor water management and inadequate access to water sanitation are a threat to the public health and the community especially those who live in remote areas, particularly the low-income households.

Rural areas that rely on untreated or poorly maintained water sources are one of those who faces global issues like water contamination. According to Nguyen et al. (2023), the quality of water in urban and rural areas depends on the connection of natural and anthropogenic factors. Stressors like farm waste runoff, domestic wastewater, climate change, and industrial wastewater, lead to the accumulation of pollutants like microorganisms which compromise the quality of water consumed on a daily basis. Rural areas are usually the most vulnerable when it comes to microbial pollution of water because they do not have adequate sanitation facilities and are usually deprived of access to proper water treatment (Nguyen et al, 2023). This means that water sources like rural hand pump wells and shallow groundwater sources are more susceptible to microbial contamination.

Structure and Components of Poso

Poso, also known as hand-pumped wells, are commonly used groundwater extraction systems in rural and peri-urban areas of the Philippines where access to piped water systems is limited. These water sources are designed to manually extract groundwater from shallow to moderately deep aquifers and serve as a

primary source of domestic water for many households (Department of Health, 2022). Contemporary research on water supply infrastructure emphasizes not only access but also the design, components, and sustainability of small-scale systems that draw groundwater using manual devices (Angelakis et al., 2025). These systems, often including hand-pumps mounted on wells, integrate mechanical and structural elements to allow reliable water withdrawal and to withstand environmental challenges.

Hand pumps are a common technology for rural water points and include key components such as the pump head, riser pipes, pump cylinder, and valves. These parts work together to lift water from beneath the water table to the surface, typically through repeated manual operation (Rural Water Supply Network, 2024). Structural integrity of these components influences how effectively the pump transfers water and how long the installation remains serviceable before repair or replacement is required.

Maintenance and functionality of hand pumps are closely tied to their physical components and structural construction. A study assessing hand pump points across multiple low- and middle-income countries showed that mechanical damage to parts such as handles and riser pipes significantly reduces the serviceability of rural water points, emphasizing that the integrity of individual components is essential to reliable performance (Li et al., 2025). In addition to mechanical parts, the well structure that supports the pump, including the casing, sanitary seal and protective platform plays a critical role in preventing

contamination and ensuring safe water delivery. International small water supply standards, such as those used in data collection and monitoring systems, define structural requirements and classification criteria for water points driven by manual pumps, including the technical specifications for installation depth, casing materials, and protections against surface water intrusion.

Although much of the current literature on hand-pump components is derived from global data, these insights are relevant to the Philippine context because *poso* systems operate on the same mechanical principles and face comparable challenges related to component durability, maintenance, and water quality protection. Understanding the design and mechanical parts of hand-pumped wells informs field-based assessments and laboratory investigations, such as those used in research to profile microorganisms entering water systems due to structural weaknesses.

In summary, recent literature highlights that a *poso's* structural effectiveness depends on the quality of components, methods of installation, and ongoing maintenance. These factors directly affect water availability, user accessibility, and susceptibility to contamination, making them important considerations in water quality studies.

Environmental Factors that affects the hand-pump

Beyond water quality, the environment in which a hand-pump operates creates a significant role in its physical durability, mechanical reliability, and functional lifespan. The World Bank (2024) review on handpump corrosion notes

that exposure to low pH or aggressive environments promotes electrochemical degradation of metals, significantly accelerating corrosion even in galvanized components. This commonly affects critical parts of the hand-pump such as pump heads, rising rods, and cylinders. Moreover, constant high humidity and moisture exposure further amplify the oxidation process, notably in tropical climates, leading to premature wear and breakdown of hand-pumps.

In addition to corrosion, other aggravations such as temperature variations, soil conditions and weather exposure considerably influence hand-pump performance and deterioration. For instance, repeated expansion and contraction caused by the temperature variations, can loosen the joints, damage seals, and crack polymer materials over time (MacAllister et al., 2022). Buried hand-pumps are affected by soil conditions such as its salinity and acidity further contributing to accelerated material degradation, misalignment and mechanical wear. In accordance with this, prolonged exposure to sunlight and rainfall degrades seals and protective coatings, increasing maintenance requirements. Studies emphasize that failure to account for these environmental conditions in the design, material selection and installation poses a significant health risk, highlighting the importance of environment specific engineering and preventive maintenance practices to ensure access to quality water for communities from hand-pumps (World Bank, 2024).

Microbial Load

Water shortage has become one of the crises that the world is currently facing due to the increase in population, changes in climate, and industrial development. According to Ryu et al. (2021), there is a demand for quality and sustainable water sources and reusing treated wastewater has become a great alternative in many parts of the world. This is recognized as a sustainable approach to provide clean water that is used in households on a daily basis. Reusing wastewater can help with the continuous demand in clean water, especially in areas with increased population. Since this has been recognized as a common practice, challenges arise in maintaining the safety of water usage, particularly due to microbial contaminants that persist despite undergoing water treatment.

Ryu et al. (2021), investigated the quality of treated water by experimenting on three anonymized wastewater treatment plants. The samples were tested for various microorganisms which included *Cryptosporidium oocysts, Giardia oocysts,* adenoviruses, bacteriophages, and aerobic endospores. Upon investigation, it was stated that water treatment did reduce the microbial load but there are still detected pathogens that persisted in the treated water. It is stated in the study that there are certain microorganisms that were more resistant to disinfection methods as it was observed that removal of protozoans were less efficient compared to bacterial and viral pathogens.

The study also highlighted that organisms that are commonly used as indicators are not always reliable which means that it does not accurately represent the quality of the water, introducing the limitations in monitoring systems of

microbial loads in water. Furthermore, the results of the study that showed persistent microbial contaminants suggests the exploration of finding a much better alternative for water treatment methods. It is also important that regulatory standards are followed in assessing the microbial safety of reused water so that a more effective water treatment can be developed (Ryu et al., 2021).

Common bacteria present in water and hand-pump

In 2023, Dong et al. examined the ability of bacteria, viruses, and protozoa in infiltration groundwater systems, posing potential health risks especially to rural areas. In the review, they explained that the contamination is largely due to both animal and domestic waste, leaking sewage systems, poor well maintenance, and runoffs from agricultural activities. In relation to that, flood-affected communities in Southern Africa have reported higher levels of *Escherichia coli* in water drawn from hand-pumped boreholes. Longitudinal monitoring following the flooding event reveals that bacterial contamination increased over time, which suggests persistent sources of pollution linked to the hand-pump environment (Rivett et al., 2022). In addition, loose hand-pump base, a cracked or damaged apron, and impaired drainage systems were found to have a strong positive relationship with increased levels of microbial contamination, suggesting compromised hand-pumps components contributed to the presence of bacteria in water supplies (Singh et al., 2021).

Biofilm formation in Water systems

Biofilm formation in water systems refers to the adhesion of microorganisms on wet surfaces, producing a protective extracellular polymeric matrix that makes these communities persistent in distribution networks and household outlets. Biofilms form in stages—attachment, maturation, and detachment—and are influenced by water quality, pipe materials, hydrodynamics, and nutrient availability (Papciak et al., 2022). For instance, Erdei-Tombor et al. (2024) say that biofilms in drinking water distribution systems are a major threat to water safety because they can hold bacteria, even chlorine-resistant strains, and lower the quality of the water by encouraging the growth of microbes on pipe surfaces. This is bad for the structure and for public health, especially for small point-of-use systems like *poso* spouts, where water that comes and goes and not much cleaning can make biofilm communities grow even more.

The biggest concern that various studies seek to resolve revolves around the means by which biofilms impair the quality of drinking water and are otherwise immune or resistant to various types of treatment and purification mechanisms. Within the context of a water mains system, Alhoshy and Samhan (2025) report that their study established that biofilms have an impactful effect on the following physical, chemical, and biological parameters in drinking water, culminating in an increase in opportunistic pathogens such as *Escherichia coli* being released in higher quantities in the flowing water. Additionally, a study by Zhu et al. (2020) involved an investigation of the biofilm potential and chlorine resistance of various strains of bacteria present in drinking water networks, prompting their study

conclusions that representative species of naturally occurring bacteria, including *Acinetobacter* and *Bacillus*, exhibited biofilms with variable biomass that approached the problem of chlorine resistance, further compounding the difficulties in ensuring its inactivation. Hence, the established fact that biofilms are a significant concern within water mains, a fact that directly relates to small-scale water outlet types such as the poso spout in Bulacan.

Although many studies, such as Erdei-Tombor et al. (2024), describe the general biofilm formation mechanism and the impact of pipe material, empirical data related to biofilm dynamics in informal water outlets remains limited. Additionally, reviews such as Oliveira et al. (2024) illustrate that drinking-water biofilms contain antibiotic resistance genes and pathogenic bacteria, but specific environmental drivers of these communities in decentralized systems are not well characterized. These gaps provide a rationale for the need for localized studies in Poso spouts, where stagnation of water, inappropriate infrastructure, and infrequent disinfection encourage unique profiles of biofilm-associated microorganisms. Thus, recommended strategies in the literature involve an integrated approach to monitoring and control: material selection, flow management, and focused disinfection supportive of the importance of bacterial isolations and profiling in the *poso* spouts in order to develop context-specific interventions for improving community water safety.

Techniques in Isolation of Bacteria

Instead of focusing on conventional water testing, the researchers aim to emphasize the need to assess microbial contaminants present in the surface and in accessible waterways of hand pumps. Biofilms, which are communities of microorganisms, tend to form on stainless-steel pipelines and other equipment surfaces (Niboucha et al., 2022). Moreover, Rao and Mulky (2023) revealed that corrosion is directly rooted from microorganisms, along with factors like water and carbon dioxide. With this in mind, poso or water spouts are closely related to the presence of biological contaminants and to corrosion, which, as a result, affects water quality. This concern calls for strict monitoring and management that ensure safe and reliable access to water, regardless of its intended use.

According to Niboucha et al. (2022), sampling methods such as swabbing, scraping, and sonic brushing were used throughout their research study in order to isolate and evaluate the formation of biofilms in stainless-steel surfaces. In line with this, swabbing involves sterile swabs soaked in phosphate-buffered saline to collect microorganisms, which are to be transferred into a bacterial suspension. Scraping utilizes a sterile spatula, most preferably flat, in scraping surfaces to remove biofilms, and is also suspended in phosphate-buffered saline. However, sonic brushing involves using an electric brush and immersing it in phosphate-buffered saline to isolate biofilms from metal surfaces. With this approach, isolation of bacteria can be applied appropriately in the research study, and the suspensions to be collected can be analyzed further by the researchers.

Bacterial Profiling

Gram Staining

Gram Staining, is the first step towards distinguishing bacteria morphology, significant in bacterial profiling. The term gram staining is derived from the bacteriologist Hans Christian Gram who discovered the staining method in 1882 (Tripathi et al, 2025). It is performed manually and is reported qualitatively, relying on visual perception to differentiate the morphology of the bacteria subjectively (Ito et al, 2023). This process involves the use of Crystal violet and Chloride ions, which can pierce through the bacterial cell wall and membranes. Crystal violet contains a positive ion that binds with negatively charged ions of the bacterial cell wall, allowing it to stain purple (Paray et al, 2023). To enhance the adherence of crystal violet, the application of iodine is optimal in this process. Iodine acts as a mordant, allowing the stain to strongly attach to the cell wall (Işıl et al., 2025).

For further differentiation and specificity, a decolorizing agent is introduced to the crystal violet-iodine complex to expose the layers of the cell membrane. Gram-positive bacteria have thick cell walls, inherently due to multiple layers of peptidoglycan layer in its cell membrane. In contrast, gram-negative bacteria contain thin layers of peptidoglycan with high lipid content in its cell wall (Paray et al, 2023).

With the addition of a decolorizing agent which targets the lipid content of the cell membranes, the gram-negative will lose its cell membrane as the lipid is dissolved due to alcohol which is commonly used as decolorizing agent. As the cell

membrane of the gram-negative microorganisms dissipate, the crystal violet-iodine complex will leak out, making the cell colorless. As for the gram-positive bacteria, the dehydration process of lipid contents will allow the cell walls to trap the crystals formed by the crystal violet-iodine complex (Işıl et al., 2025).

Following the decolorization process will be counter-staining where a weaker stain namely Safranin is introduced to the cell wall of the bacteria. This stain will allow the gram-negative to be visualized for its cell morphology. Safranin does not mask the previous crystal violet in Gram-positive bacteria as the crystal violet is bound by the iodine to the cell wall (Paray et al, 2023).

These are the processes that make up the gram staining technique. However, it is important to note that although through this methodology, bacteria can be classified by morphology, from its shape and through its color; it also requires observational bias and further experience. As stated in the study of Yamaji et al (2021), specimens such as *Pseudomonas aeruginosa* and *Enterobacteriaceae* may be perceived as one another based on their morphology. This mistake may result in multiple unnecessary antibiotics. Subsequently, although Gram stain can be a stand alone procedure for screening, it usually follows additional laboratory testing for more specificity.

Eosin Methylene Blue

Eosin Methylene Blue (EMB) agar, a differential and selective medium, is commonly used for establishing the presence of enteric bacteria, particularly *Escherichia coli*. The medium contains dye, namely eosin and methylene blue,

which function both as selective agents as well as indicators for lactose fermentation. These dyes inhibit the growth of Gram-positive bacteria organisms, thereby enhancing the isolation of Gram-negative bacteria. Furthermore, a study conducted by Lestari, Haliza, and Setiawan (2024) states that methylene blue plays a significant role in suppressing the growth of Gram-positive bacteria, contributing to the selectivity of the medium. With regards to differentiation, lactose-fermenting microorganisms have the ability to produce acid that reacts with the dyes, resulting in distinct colony coloration. Strong lactose fermenters, such as *E. coli*, typically exhibit a dark-colored colony with its characteristic metallic green sheen, while non-lactose fermenters produce colorless to pale colonies. Additionally, variation in environmental conditions, such as pH, may influence bacterial growth and colony appearance on EMB agar, which can potentially influence the accuracy of interpretation.

Biochemical tests

Pathogens produce specific metabolites in response to chemical triggers, including contact with pharmaceuticals and other factors. Moreover, these pathogenic strains are often associated with diseases found in humans and animals, making proper identification essential in order to develop effective treatments and growth interventions. Enzymes catalyze biochemical reactions which then allows bacteria to grow, reproduce, and respond to conditions. Due to this factor, the enzymatic activities of the pathogens may be used to identify and characterize specific bacterial isolates (Hafezi & Khamar, 2024).

Catalase Test

Catalase is an enzyme composed of porphyrin groups which allows for its reaction with hydrogen peroxide. This test is essential in differentiating *Streptococcus* and *Staphylococcus* bacteria, which are facultatively anaerobic and aerobic bacteria, respectively. Hydrogen peroxide stimulates the release of oxygen gas when it comes into contact with catalase, appearing as visible bubbles and protecting the bacteria from toxicity and cellular damage (Hafezi & Khamar, 2024).

Coagulase Test

The coagulase test is used to differentiate *Staphylococcus aureus* and coagulase-negative staphylococci (CoNS). The production of coagulase enzyme by *Staphylococcus aureus* is evaluated through this test. The production of this enzyme leads to formation of clot in the plasma due to the conversion of fibrinogen to fibrin. The test can be performed in a slide or tube method. The tube method mixes the human plasma collected from the Ethylenediaminetetraacetic acid tube and the bacterial colony that is tested for the presence of S. aureus bacteria. Clotting of the plasma in the test tube is recorded as positive for coagulase enzyme production and confirms the presence of *Staphylococcus aureus*. (Choudhary et al., 2022).

Oxidase Test

Oxidase are enzymes that are found in the intracellular system of pathogens containing cytochrome-c. This intracellular system catalyses the transport of electrons which then reacts with oxidizes with the integration of Kovac's reagent,

containing 1% tetra-methyl-*p*-phenylenediamine dihydrochloride. An oxidase positive reagent forms a colored compound, producing blue or purple. This test is used to determine bacteria species such as *Neisseria, Moraxella,* and *Campylobacter* (Chavan et al., 2022).

Urease Test

The urease test is a method used in the detection of *Helicobacter pylori.* This bacteria typically uses urease to catalyze the hydrolysis of urea that allows it to turn into ammonia and carbon dioxide (Cui et al., 2022). The production of ammonia raises the pH level of the agar, which can be used as a pH indicator. This test is rapid and inexpensive, making it practical for identification of urease-producing bacteria (Hibaoui et al., 2022).

SIM Test

Sulfide Indole Motility agar is usually utilized in determining bacterial motility. This semi-solid medium becomes turbid along the stab line when in contact with motile-positive bacteria, indicating their ability to move away from the inoculation point Moreover, the agar may also be used to determine the pattern of the bacterial growth, whether it spreads vertically or horizontally. The medium also allows for the detection of hydrogen sulfide production, which reacts with ferric ions, creating black precipitates (Hafezi & Khamar, 2024).

Simmons Citrate Test

Simmons Citrate Test is a method derived from the use of citrate salt, typically used as a carbon source for bacteria. Bacteria that synthesize enzymes

citratease, citrase, and citrate lyase make use of the citrate in order to actively perform metabolic functions. Additionally, it uses a pH indicator, bromothymol blue, that turns from green at neutral pH to a blue color in the presence of a citrate-positive bacteria (Hafezi & Khamar, 2024).

Methyl Red and Voges-Proskauer Tests

This method is conducted with the use of glucose fermentation pathways to identify facultative anaerobic enteric bacteria. For the Methyl Red test, with the addition of methyl red as indicator, the results affirm a positive reaction when there is persistent red tone appearance in the upper portion of the medium. This is because Methyl red detects if the bacteria uses a mixed acid fermentation pathway whilst doing the test. The bacteria produce organic acids from this pathway and is indicated when the pH of the medium falls to its acidic phase, turning the surface to red. Meanwhile, the medium will turn into yellow if it detects an alkaline condition as it produces neutral end products (Aryal, 2018, as cited in Anokhe, Kalia, & Shanmugaraj, 2021).

On the other hand, Voges-Proskauer Test is performed to determine the neutral end result of glucose fermentation, taking note of its acetoin content. This test is used to detect the production of butanediol instead of organic acids. Following this, the addition of 40% KOH and atmospheric oxygen, acetoin becomes diacetyl. Moreover, alpha-naphthol is added to determine the presence of diacetyl turning it into a brownish red-complex, indicating a positive result (Aryal, 2018, as cited in Anokhe, Kalia, & Shanmugaraj, 2021).

Indole Test

This test is undertaken to detect *Enterobacteriaceae family*, including fecal indicator *Escherichia coli*. The indole test targets the enzyme tryptophanase, which is found in bacteria allowing it to break down tryptophan found in the medium and excrete indole gas. Moreover, tryptone broth, incubated in 37ºC for 24 hours, is recommended for this test as it contains tryptophan as substrate. Addition of Kovac's or Ehrlich-Boehme's reagent to the surface allows for the detection of indole gas, indicated by a red ring seen 2 minutes after adding (Hafezi & Khamar, 2024).

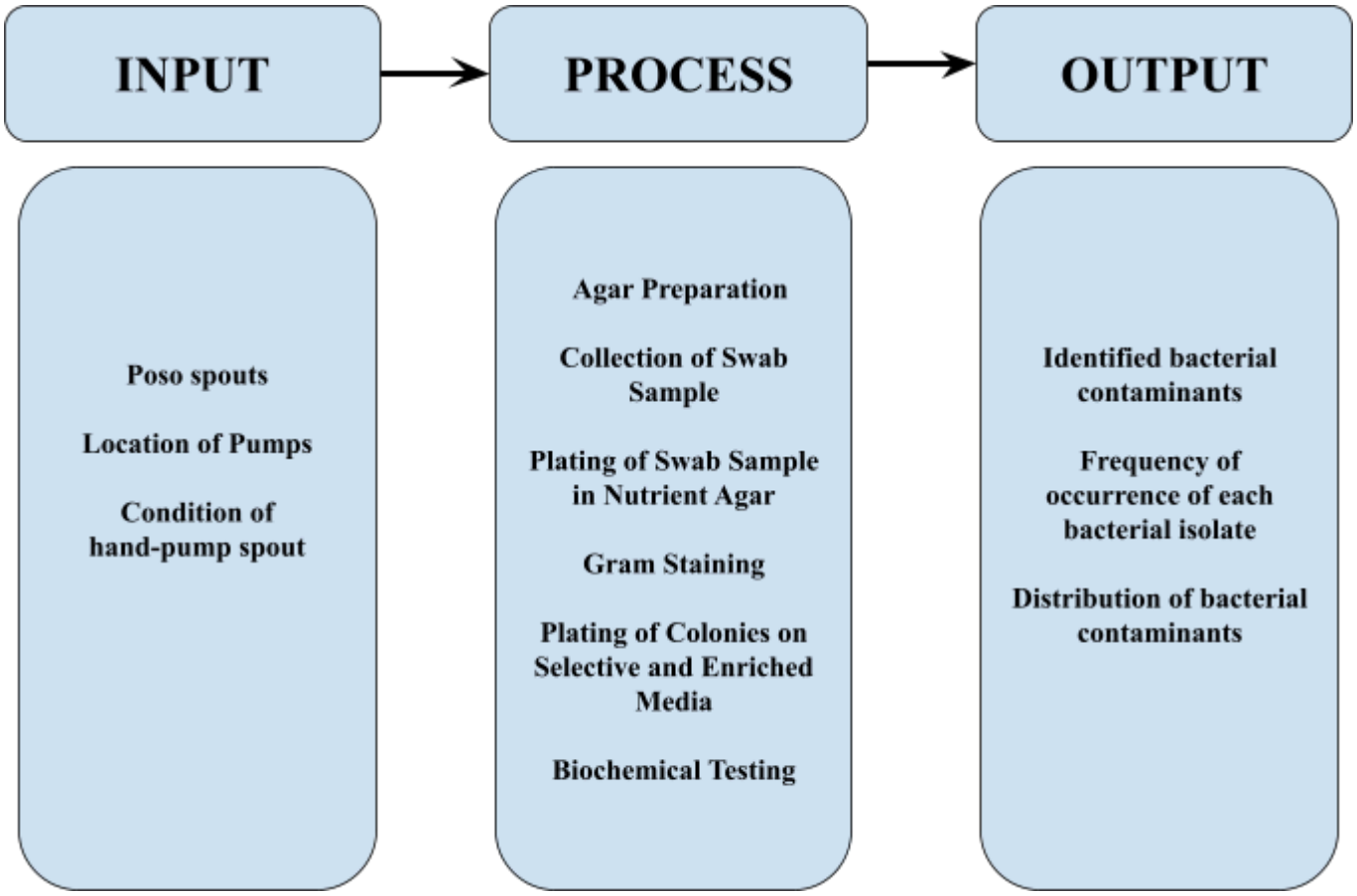
Triple Sugar Iron Test

Triple Sugar Iron Test involves a variety of concentration of sugars in the culture medium which includes 1% lactose, 1% sucrose, and 0.1% glucose. These concentrations allow for the assessment of the bacteria's capability to ferment the glucose. This test is primarily utilized to detect *Enterobacteriaceae* family and to differentiate them to other gram-negative bacteria. Moreover, it also includes the ph indicator, Phenol red which detects the acidity of the medium. This indicator allows for the color change of the medium turning it from yellow to red after acid production. The medium is also capable of determining the gas production of H2S in the form of colored precipitate. This occurs due to the presence of sodium thiosulfate and iron sulfate, which are respectively the substrate for the H2S production and the color reagent (Hafezi & Khamar, 2024). The expected outcome of this involves the following: Slope and Butt color that indicates the carbohydrate

fermentation and pH of the medium. Red slope and yellow butt color indicates the glucose fermentation where it is produced under low oxygen conditions while the slope remains alkaline as it utilizes peptone compounds instead of glucose, which produces ammonia making it alkaline. Yellow slope and butt indicates lactose or sucrose fermentation with acid production. Lastly, Red slope and butt indicates that the bacteria is a non-carbohydrate fermenter and there is no production of acid, H2S gas or gas. The red color is indicative of the alkaline environment as there is peptone consumption instead of glucose. The separation or cracking of culture medium is also observed in these tests to monitor the production of gas. Lastly, blackening of the medium indicates the presence of H2S gas production due to the reduction of sodium thiosulfate(Hafezi & Khamar, 2024).

Conceptual Framework

Figure 1.
Conceptual Framework



A conceptual framework establishes the foundation for addressing the relationship between the main concepts of the study and variables involved. It introduces and expounds on factors illustrated through written descriptions and visual representations (Luft et al, 2024). According to Salawu et al (2023), it is the overall ideal conceptualization of the researchers, allowing organizations of approaches and desires of how the study will progress. The conceptual framework (Fig. 1) that will be used in the study adapts the Input-Process-Output approach or IPO model to illustrate the flow of the variables involved in the bacterial isolation and profiling of hand-pump spouts in Bulacan.

The input variables represent the factors that will directly influence the experimental process. These include location of hand-pumps, specific condition of the hand-pump spout, such as the presence of rust, damage, cracks, and biofilm formation, and the overall condition of the hand-pump system, such as damaged aprons, loose pump bases, poor drainage around the pump area.

The process component describes the laboratory and analytical procedures conducted to attain the objectives of the study. The process will begin with securing site access permission from barangay officials and hand-pump owners. No human participants or personal data will be involved, as the study is limited to environmental sampling of hand-pump spouts. This will be followed by direct swab collection using sterile cotton swab from the hand-pumps spout. Following collection, the collected swabs will be properly stored in Amies transport medium and stored at 2–8°C in a cooler during transport to the laboratory for analysis.

Isolation of bacterial contaminants will then be performed using appropriate culture media. Subsequently, bacterial profiling will be conducted through colony morphology observation, gram staining, subculturing on selective and enriched medias and standard biochemical assays.

The output in this model refers to the expected findings of the experiment, which include identified bacterial contaminants isolated from hand-pump spouts, frequency of occurrence of each bacterial isolate, and the distribution of bacterial contaminants among selected hand-pumps locations in Bulacan. The findings obtained from this framework will serve as the evidential basis for assessing public health risks and recommending improvements in sanitation practices.

Definition of Terms

The following are terms used by the researchers in the study:

Colony-Forming Units (/mL). A standard unit of measurement in microbiology used to quantify the viable number of living microbial cells (or clusters of cells) per milliliter of water, serving as the basis for measuring the reduction in microbial load.

Hand-Pump Water Source. It is a non-electric, manually-operated well pumping device that uses human power to acquire water from the water ground. It is mostly found in rural areas where water source is lacking, therefore serving as a primary water source for consumption and domestic use.

Microbial Contaminants/Load. The total viable population of microorganisms present in the water sample, which is the target variable whose initial presence and subsequent reduction are quantified using /mL.

CHAPTER 3

Methodology

Research Design

The study will employ a descriptive analytical research design, which focuses on collecting and analyzing data to determine measurable outcomes. This approach allows researchers to objectively assess changes through the use of statistical tools, and identify, and characterize the microorganisms present in poso spouts through the use of Gram staining and biochemical tests aligned with the objectives of the experiment. The researchers will use purposive sampling, targeting hand-pumps that are used in the community and privately-owned by the residents while excluding those not utilized by them. The research study employs environmental sampling, specifically the swab collection from the spouts of the selected poso. The study does not involve any human participation nor require the collection of any personal information. As part of the selection process, the posos will be checked for their observable environmental conditions such as presence of rust and debris, hygiene of the surrounding, and signs that the posos have been used.

Research Instrument

The research instrument that the group will use is an observational and experimental approach on the evaluation of the isolated bacteria from the swab samples collected from six selected hand-pumps of the barangays in Bulacan. Sterile swabs will be collected directly from the spouts of the hand pumps

commonly used by the community in order to identify and evaluate the characteristics of bacterial microorganisms present. In order to ensure ethical compliance, a letter of request for permission will be provided and submitted to the concerned local government unit and the respective water pump owners during the process of selection of hand pumps for this study. The aforementioned documents will state the researchers' purpose and procedures in conducting the study regarding water hand pumps, and will serve as a formal request for permission to utilize the poso as the sampling site for collection and analysis of the microorganisms that may be present on its spouts for bacterial profiling.

The research instruments used in this study include sterile cotton swabs and transport media for sample collection and ensure bacterial integrity; inoculating loops, glass slides, nutrient and selective agar plates for bacterial culture and isolation; Gram staining reagents; biochemical test reagents such as 3% hydrogen peroxide, oxidase reagent, Methyl Red reagent, and other specific reagents required in tests appropriate for bacterial identification. A microscope will be used for microscopic observation, and description will be used. Statistical analysis of the collected data will be performed using Jamovi and Excel software.

Validation of Research Instrument

For better understanding and interpretation of the results, two (2) registered medical technologists will be consulted to ensure the accuracy and proper identification of microorganisms found in the samples. Furthermore, the researchers will also be guided by the quality control standards established by

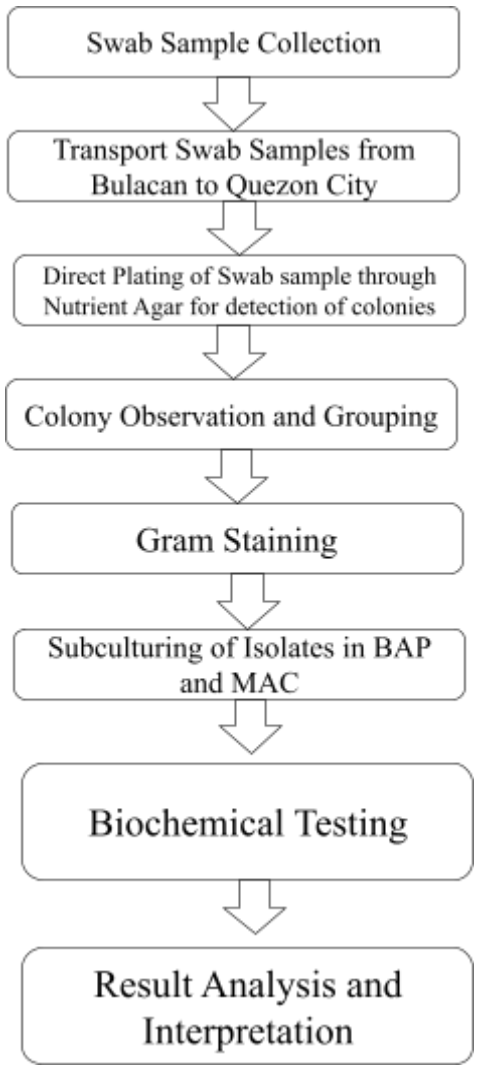
CLSI and WHO to ensure methodological quality and suitability of the laboratory procedures to be done in order to obtain the results.

All materials and instruments requiring incubation will be prepared in advance to prevent cross-contamination and maintain sterility throughout the entire laboratory process. Aseptic techniques will also be practiced to ensure the reliability and validity of the procedure.

Experimentation Proper

Figure 2.

Experimentation Proper



Research Locale

This study will be conducted in selected barangays that have hand-pumped sources for community and household use, a structure common in rural areas across

the province. The selected barangays will serve as representatives for microbial examination in rural communities, especially those remotely accessed.

Selection of Barangay

The barangays selected are based on the reliance of the community on hand-pumped groundwater systems. The barangays chosen for study sites are based on the number of poso sites found available in Bulacan. The continued utilization of these hand pump systems makes the barangays the best choice for examining the sanitation of community water sources in a rural context. The decision to choose these barangays are also based on the researcher's accessibility conditions and the research's approach and objectives. Additional information for these barangays will remain anonymous to ensure ethical considerations towards the resident and prevent negative publicity to the barangays.

Selection of Hand Pumps

The selection of hand pumps following six hand-pumped groundwater sources will be selected as the sampling site prior to data collection for its accessibility, and their role as a water source within the barangays. The hand-pumps will undergo a selection criteria based on observable environmental conditions such as presence of rust and debris, hygiene of the surrounding, and active usage of the posos to determine significant microbial findings. Since the community's level of utilization will be taken into consideration as an important criterion for selection of sampling sites, statistical data coming from the barangays will be requested.

Data Collection Procedures

Gathering Consent

Prior to the actual data collection, a letter of request will be sent to the local government units and owners of the households in the selected barangays in Bulacan to obtain permission for the conduct of the study. To guarantee that the residents and barangays, who are responsible for the authorization of the posos, completely comprehend the study's aim, objectives, and processes, a thorough explanation will be provided accordingly. Additionally, the prospective owners will be apprised of their rights, which include the assurance that the researchers will uphold their identities with utmost confidentiality and anonymity, the option to withdraw at any time during the process, and voluntary. The following barangays will not be disclosed to the public to respect the identity and reputation of the barangays and the owner of the posos. Following this, there will be no database that will show the coordinates of the locations of sampling. For presentation purposes, each site will be assigned a serial code that is not directly linked to the locations. The posos will be displayed under non-identifiable codes such as "Poso A", "Poso B" and so on.

Agar Preparation

Before sampling, the agar media for the respective tests will be prepared to ensure immediate inoculation and bacterial isolation following specimen collection.The preparation of agar media will be adapted from the manufacturer instructions, specifically those provided by HiMedia Laboratories and TM media.

Appropriate conversion and measurement of volume of distilled water and agar powder base will also be done to ensure accurate media formulation and consistency.

Nutrient Agar

For Nutrient Agar, the researchers will suspend 9.24 grams in 330 mL distilled water. The mixture will be heated to boiling to dissolve the medium completely. To achieve sterility, the researchers will autoclave the medium at 15 lbs pressure (121ºC) for 15 minutes. It will then be cooled to 45-50ºC, mixed well and will be poured aseptically into sterile 18 Petri plates. 15 mL of Nutrient Agar will be poured per plate. A control will also be put in incubation for quality assurance. Presence of microorganisms in the agar will indicate contamination and unsuccessful sterilization, resulting in the need for repeat preparation of agar media.

MacConkey Agar Plate

For MacConkey Agar Plate, the researchers will suspend 16.4 grams of MacConkey Agar powder base in 330 mL purified distilled water. The mixture will be heated to boiling to dissolve the medium completely. To achieve sterility, the researchers will autoclave the medium at 15 lbs pressure (121ºC) for 15 minutes. It will then be cooled to 45-50ºC, mixed well and will be poured aseptically into sterile 18 Petri plates. 15 mL of MacConkey Salt Agar will be poured per plate. A control will also be put in incubation for quality assurance. Presence of microorganisms in the agar will indicate contamination and unsuccessful

sterilization, resulting in the need for repeat preparation of agar media.

Eosin Methylene Blue Agar Plate

For Eosin Methylene Blue Plate, the researchers will suspend

Blood Agar Plate

For Blood Agar Plate, the researchers will suspend 1.32 grams of Blood Agar powder base in 330 mL purified distilled water. The mixture will be heated to boiling to dissolve the medium completely. To achieve sterility, the researchers will autoclave the medium at 15 lbs pressure (121ºC) for 15 minutes. It will then be cooled to 45-50ºC, mixed with 5% w/v sterile defibrinated blood and will be poured aseptically into sterile 18 Petri plates. 15 mL of Blood Agar Plate will be poured per plate. A control will also be put in incubation for quality assurance. Presence of microorganisms in the agar will indicate contamination and unsuccessful sterilization, resulting in the need for repeat preparation of agar media.

Mannitol Salt Agar

For Mannitol Salt Agar, the researchers will suspend 36.7 grams of Mannitol Salt Agar powder base in 330 mL purified distilled water. The mixture will be heated to boiling to dissolve the medium completely. To achieve sterility, the researchers will autoclave the medium at 15 lbs pressure (121ºC) for 15 minutes. It will then be cooled to 45-50ºC, mixed well and will be poured aseptically into sterile 18 Petri plates. 15 mL of Mannitol Salt Agar will be poured per plate. A control will also be put in incubation for quality assurance. Presence of

microorganisms in the agar will indicate contamination and unsuccessful sterilization, resulting in the need for repeat preparation of agar media.

Tryptone Broth

For Tryptone Broth, the researchers will dissolve 1.65 grams of tryptone broth agar base in 110 mL distilled water. The mixture will be heated to boiling to dissolve the medium completely. To achieve sterility, the researchers will autoclave the medium at 15 lbs pressure (121ºC) for 15 minutes. It will then be cooled to 45-50ºC, mixed well and will be poured aseptically into sterile 18 tubes with screw caps. 5 mL of Tryptone Broth will be poured per tube. A control will also be put in incubation for quality assurance. Presence of microorganisms in the agar will indicate contamination and unsuccessful sterilization, resulting in the need for repeat preparation of agar media.

Sulfide Indole Motility

For Sulfide Indole Motility Agar, the researchers will suspend 3.993 grams of SIM agar powder base in 110 mL distilled water. The mixture will be heated to boiling to dissolve the medium completely. It will be mixed and dispensed aseptically into sterile 18 tubes with screw caps. 5 mL of Sulfide Indole Motility Agar will be poured per tube. To achieve sterility, the researchers will autoclave the tube medium with the screw cap partially closed at 15 lbs pressure (121ºC) for 15 minutes. It will then be cooled to 45-50ºC in an upright position. A control will also be put in incubation for quality assurance. Presence of microorganisms in the agar will indicate contamination and unsuccessful sterilization, resulting in the

need for repeat preparation of agar media.

Simmon's Citrate Agar

For Simmon's Citrate Agar, the researchers will suspend 2.67 grams of SCA powder base in 110 mL distilled water. The mixture will be heated to boiling to dissolve the medium completely. It will then be mixed and dispensed aseptically into sterile 18 tubes with screw caps. 5 mL of Simmon's Citrate Agar will be poured per tube. To achieve sterility, the researchers will autoclave the tube medium with the screw cap partially closed at 15 lbs pressure (121ºC) for 15 minutes. It will then be cooled to 45-50ºC in a slant position. A control will also be put in incubation for quality assurance. Presence of microorganisms in the agar will indicate contamination and unsuccessful sterilization, resulting in the need for repeat preparation of agar media.

MR-VP Broth

For MR-VP broth, the researchers will suspend 1.87 grams of MR-VP powder base in 110 mL distilled water. The mixture will be heated to boiling to dissolve the medium completely. It will then be dispensed aseptically into sterile 18 tubes with screw caps. 5 mL of MR-VP broth will be poured per tube. To achieve sterility, the researchers will autoclave the tube medium with the screw cap partially closed at 15 lbs pressure (121ºC) for 15 minutes. It will then be cooled to 45-50ºC. A control will also be put in incubation for quality assurance. Presence of microorganisms in the agar will indicate contamination and unsuccessful

sterilization, resulting in the need for repeat preparation of agar media.

Triple Sugar Iron Agar

For Triple Sugar Iron Agar, the researchers will suspend 7.1 grams of TSI powder base in 110 mL distilled water. The mixture will be heated to boiling to dissolve the medium completely. It will then be mixed and dispensed aseptically into sterile 18 tubes with screw caps. 5 mL of Triple Sugar Iron Agar will be poured per tube To achieve sterility, the researchers will autoclave the tube medium with the screw cap partially closed at 15 lbs pressure (121ºC) for 15 minutes. It will then be cooled to 45-50ºC and set in a sloped form with a butt about 1 inch long. A control will also be put in incubation for quality assurance. Presence of microorganisms in the agar will indicate contamination and unsuccessful sterilization, resulting in the need for repeat preparation of agar media.

Collection of Swab Sample

Swab samples will be collected from selected hand pumps in the barangays in Bulacan. Before collection, the people around the area will be informed of the collection and the risks. They will be advised to leave the vicinity before and during collection to ensure their safety. The sample collection guidelines from swabbing hand spouts will follow standard microbiological protocols and procedures from Centers for Disease Control and Prevention (2024). The sampling, as stated in the study of Stocker, Smith, and Pachepsky (2022), will be performed in the morning, as hand-pumps typically exhibit higher bacterial count during this

time of the day, ensuring that the samples retrieved will reflect peak microbial contamination. To prevent sample contamination, the sampling process will strictly comply with aseptic techniques to ensure that the sample is in optimal condition.

The swab procedure will follow with considerations of the microbe recovery provided by United States Pharmacopeia (2025) and Research SOP (2023). Prior to sample collection, the poso spout will undergo disinfection to ensure the removal of external microbes or contaminants. This will be done using 70% isopropyl alcohol. The same disinfection procedure will be done after sample collection to ensure that there are no microorganisms left on the surface. Personal protective equipment will be utilized, specifically laboratory gown, disposable nitrile gloves, hair cap, N95 face mask, shoe cover, and face shield. First, the researchers will check for the surface moisture condition, ensuring that the surface is dry as wet contact surface may result in a low percentage of recovering microorganisms. The surface of the poso spout will be swabbed in a close parallel pattern with consistent pressure. The swab will be held at a 30º angle to allow for maximum contact between the cotton swab and poso spout surface. The expected area to be covered per swab will be 25-30 cm^2. The duration of the procedure will be about 5-10 minutes including the preparation.

Each swab sample will be collected in a sterile, tightly sealed container containing Amies transport medium that will be properly labeled with necessary information such as the date, time, source location, and sampler information. In addition, the researchers will ensure that when sampling, the interior surfaces of the

container and lid are not contacted (PathWest, 2021). After collection, the samples will be stored immediately in a cooler with ice and maintained at 2-8°C during transport to preserve the quality and microbial integrity, as recommended by PathWest and Infanta Water District. To prevent cross contamination, triple packaging will be implemented. The transportation vehicle to be used for the delivery of the samples from Bulacan to Trinity University of Asia will be carried out using the researchers' personal vehicle to minimize risk of contamination and ensure proper specimen handling conditions.

The collected samples will be delivered to the microbiology laboratory of Trinity University of Asia within 3 hours, a recommended and generally recognized period of holding time to avoid variations in the microbial load (PathWest, 2021; Infanta Water District, 2024). Upon arrival at the laboratory, the swab samples will promptly undergo microbiological analysis to establish the presence or absence of bacterial contaminants, and the identity of the detected bacteria.

Plating of Swab Sample in Nutrient Agar

Procedural methods will be done to identify the microorganisms present in the sample. Direct microscopy is one of many methods that can be used to observe the live cell microorganisms through differential staining techniques. However, this study uses the direct agar-plated methods that have been developed to reduce the volume of reagents and media used. It is a straightforward technique that is used to screen microbes that thrive at moderate temperatures.

Spread plate agar method will be performed in order to ensure that even small amounts of bacteria will be included. First, the researchers will remove the swab from the Amies transport medium and insert it on a test tube containing normal saline solution. This is to allow the microorganisms to spread evenly. Following this, a small sample of 0.1mL will be aseptically transferred and spread evenly over the volume onto the center of the agar plate to the surface of the medium using either a cell spreader or a bent glass rod that will be flamed sterile over the bunsen burner at an angle of 45 degrees. The spreader must move in either circular path or in a back and forth motion for spreading evenly and easily. Lastly, the plate will be left in a flat position to allow the sample to absorb into the agar for at least 3-5 minutes after spreading.

After 3-5 minutes, the plates will be incubated at an appropriate temperature, commonly 37°C for 24-48 hours for bacterial growth. This will allow the organisms to properly grow and form colonies before assessment. For screening, after incubation, the appearance of spots will be referred to as an indication of positive growth of certain bacteria. Growth patterns, colony morphology, and viability must be observed and quantified for better understanding.

This procedure will be repeatedly done for all 6 swab samples that were collected in the hand-pumps of representative barangays in Bulacan.

Gram Staining

Gram staining will be used to verify that presumptive colonies are Gram-negative. On a spotless glass slide, a tiny smear of the colony will be made, allowed to air dry, and then heat-fixed. Crystal violet (1 min.), Gram's iodine (1 min.), ethanol decolorization, and safranin (1 min.) counterstaining will be applied to the smear in that order. The slide will be examined under oil immersion once it has been blotted dry. Expected result: short pink or red rods, indicating Gram-negative bacteria.

For the Gram-staining process, it will require clean glass slides, an inoculating loop, a Bunsen burner for heat fixation, distilled water, and the four staining reagents: crystal violet as the primary dye, Gram's iodine as the mordant, acetone alcohol as decolorizer, and safranin as the counterstain.

To begin the procedure, a small amount of bacterial culture will be spread thinly on a glass slide using a sterile loop and is allowed to dry in air. Once dry, the smear will briefly pass over a flame to secure the cells on the slide. Crystal violet will then be applied to cover the smear and allowed to sit for about one minute before being rinsed gently with distilled water. This will then be followed by the addition of Gram's iodine, which will remain on the slide for another minute to enhance dye binding. After rinsing again with distilled water, the smear will be exposed to acetone alcohol, which will remove the crystal violet-iodine complex from bacteria that do not retain the dye. The decolorization step will be kept brief

to prevent over-decolorizing, and the slide will again be washed with distilled water. Safranin will then be added for approximately one minute to provide a contrasting color to cells that lost the primary stain. After the final rinse, the slide will be left to air-dry and will be examined under the microscope.

In this staining method, Gram-positive bacteria will appear purple because they retain the crystal violet-iodine complex, while Gram-negative bacteria will take up the safranin counterstain and appear pink or red.

Plating of Colonies on Selective and Enriched Media

MacConkey Agar Plate

The MacConkey Agar Plate will be used as a selective and differential medium to isolate Gram-negative bacteria and to differentiate them based on lactose fermentation. MacConkey Agar preparation will include MacConkey agar culture medium. The bacterial isolate will be inoculated onto the surface of the MacConkey Agar Plate using the 4-quadrant streaking technique and obtain isolated colonies and will be inoculated at 37°C for 18-24 hours. After incubation, the plate will be observed for bacterial growth and lactose formation. Positive results will be indicated by the formation of pink to red colonies, which signify lactose fermentation, while non-lactose fermenters will produce colorless colonies.

Blood Agar Plate

The Blood Agar Plate will be used as an enriched and differential medium to support the growth of a wide range of bacteria and to determine their hemolytic activity. Blood Agar Plate preparation will include blood agar medium. The bacterial isolate will be inoculated onto the surface of the Blood Agar Plate using the 4-quadrant streaking technique to obtain isolated colonies and will be incubated at 37°C for 18-24 hours. After incubation, the plate will be examined for bacterial growth and hemolytic reactions. Positive results will be indicated by hemolysis surrounding the colonies, such as beta-hemolysis, characterized by clear zones, alpha-hemolysis showing greenish discoloration, and gamma-hemolysis indicating no change in the medium.

Mannitol Salt Fermentation Test

Mannitol Salt Fermentation test will include the following: mannitol salt agar culture medium and bacterial isolate. Unknown bacteria will be isolated and streaked across mannitol salt agar using 4-quadrant streaking technique. Incubation will be done to the culture medium at 37°C for 18-24 hours. It will then be observed for bacterial growth and color changes. A color change from red to yellow may indicate mannitol salt fermentation. Note that this targets microorganisms that are salt-tolerant. Hence, positive results will indicate its presence. No color change may indicate absence of salt-tolerant bacterial growth.

Triple Sugar Iron (TSI) Test

The Triple Sugar Iron (TSI) Test will be performed using TSI agar, an inoculating needle, and test tubes. The agar will be prepared, dispensed into tubes, and allowed to solidify in a slanted position. The bacterial sample will be then inoculated by streaking the slant surface in a zigzag pattern and stabbing the butt of the medium. The tubes will be incubated at 37°C for 18-24 hours. After incubation, the slant and butt are examined for color changes. A positive result will be indicated by acid production, seen as a yellow slant and yellow butt, with black precipitate and no cracks. A negative result will show alkaline conditions with a red slant and red butt, no black precipitate, and the presence of cracks.

IMViC Test

Indole test

The Indole test requires test tubes, inoculating loops, and tryptone broth. A tube containing 1% tryptophan is inoculated with a few colonies of the bacterial culture and will be incubated at 37°C for 24 hours. After incubation, Kovac's reagent or the Ehrlich-Boehme's reagent will be added to the surface of the broth. A positive result is indicated by the formation of a red ring at the surface, while a negative result will show no color change or a yellow to reddish-reddish brown color.

Methyl Red Test

For the Methyl Red test, Methyl Red/ Voges-Proskauer broth, an inoculating loop, and Methyl Red reagent will be used. The broth will be

inoculated with pure bacterial colonies and incubated at 37°C for 48-72 hours. After incubation, 5 drops of Methyl Red reagent are added to the broth. A positive result is confirmed when the broth turns red, whereas a negative result is shown when the broth turns yellow.

Voges-Proskauer (VP) Test

Voges-Proskauer (VP) test also uses Methyl Red/Voges-Proskauer broth, an inoculating loop, and VP reagent. The broth will be inoculated with pure colonies and incubated at 35-37°C for 24-48 hours. Following incubation, Barritt's reagents A and B are added in a 3:1 ratio and mixed. A positive result is indicated by the development of a red or pink color, while a negative result shows no color change.

Citrate Utilization Test

The Citrate Utilization test is conducted using Simmon's Citrate Agar and an inoculating loop. The agar will be inoculated with a pure bacterial culture and incubated at 37°C for 48-72 hours. A positive result is observed when there is visible growth on the medium along with color change from green to blue. A negative result shows no growth and no change in color.

The following tests were adapted from Banar et al. (2024) with minimal modification in accordance with the biosafety guidelines, following aseptic technique and standard laboratory practices for handling microorganisms.

Catalase Test

The catalase test will require glass slides, hydrogen peroxide solution, swab or non-metallic ounce, and the bacterial isolates. A clean glass slide will be labeled

with Add a drop of 3% hydrogen peroxide solution on a glass slide. Inoculate a small amount of bacterial colony using a swab or non-metallic ounce and mix with the hydrogen peroxide. Observe the mixture. Positive test: formation of bubbles indicate presence of Staphylococcus species. Negative result: no bubble formation indicates presence of Streptococcus species

Coagulase Test

The coagulase test is used to differentiate the different strains of *Staphylococcus aureus* from *Staphylococcus epidermidis* and also from coagulase-negative staphylococci (CoNS). The test can be done with two types of preparation, the tube and slide method. The tube method is performed by mixing the plasma collected with EDTA tube and the isolated colonies using a sterile loop. The test is checked for visible clotting after it is incubated at 37 for 18-24 hours for confirmation. After incubation, the tube will show visible clotting indicating a positive result for free coagulase, which confirms the presence of *Staphylococcus aureus*.

Oxidase Test

To confirm the oxidase activity, presumptive colonies from Nutrient Agar will be subjected to the Oxidase test. A sterile loop will be used to transfer a colony after a drop of oxidase reagent is deposited on filter paper. A positive outcome will be indicated by a quick change in color to purple or dark blue within 10 to 30 seconds.

Motility Test (SIM or Motility Agar)

Test for Motility (SIM or Motility Agar) SIM medium or semi-solid motility agar will be used to assess bacterial motility. The colony will be stabbed directly into the center of the medium using an inoculating needle, and it will be cultured for 24 to 48 hours at 35 to 37°C. The motility of the bacteria will be observed based on the diffusion of the growth away from the stab line which indicates its movement.

Statistical Analysis

This study will employ descriptive analytical methods to evaluate the microbial quality of the poso spouts samples. All analysis will be processed using Excel and Jamovi, which offers accessible tools for statistical testing and data visualization.

To begin, descriptive statistics including frequency counts and percentages will be used to summarize the presence of bacterial contaminants. This will include a summary of findings regarding specific bacteria types, their Gram-staining status, and biochemical results. Furthermore, the bacterial distribution and types found in each poso will be calculated.

A descriptive analysis will be carried out, but only after bacterial growth has been assessed through agar culturing. At this stage, the samples will be

examined for visible physical changes, including color shifts, cloudiness, and the presence of particulate matter.

To present the results clearly, tables will be used for summarizing the findings and microbial diversity. Bar charts will be used to visualize the type of bacteria present in each hand-pump, it serves as a primary tool for pattern recognition. These illustrations will facilitate the interpretation of whether specific microbes represent random isolated occurrences or consistent contamination patterns across multiple collection sites. Line graphs will also be used for microbial load comparisons and ANOVA results.

Ethical Considerations

Before commencement of the study, and as part of institutional ethical requirements, the research proposal will be reviewed and evaluated by the Institutional Ethics Review Committee of the Trinity University of Asia. Approval from both the International Ethics Committee and the Biosafety and Security Committee (BSC) shall be obtained prior to any experimentation. These approvals are required before any experimentation can proceed and ensure that the study complies with ethical, legal, and biosafety standards.

In collecting swab samples from hand pumps in selected barangays of Bulacan to test their microbial contamination, it is crucial to integrate ethical and moral principles throughout the process. Thus, the researchers will obtain

permission from local residents and community leaders through a letter of request to ensure that swab collection from the hand pumps respects community ownership and local governance structures. Furthermore, the study proponents will ensure that the results benefit the local community, by identifying contaminations and its sources through the results of the study thereby contributing to safer water practices or sustainable water treatment solutions accessible for all.

The study proponents shall practice preventive and control measures to ensure public safety. The following shall be practiced: researchers shall avoid excessive sampling that could disrupt access or quality of suspension; strictly follow field hygiene practices, such as using washing of hands prior to sampling; ensure the use of complete personal protective equipment, and disinfection of sampling points to avoid contamination of the contact source. Additionally, safe working distance must be set in place during sampling to ensure the safety of the surrounding environment and individuals present, and conduct all procedures in a manner that does not cause disruption risk.

Biosafety and biosecurity principles will also be strictly followed before, during, and after the processing of both bacterial suspension. Prior to transport, all materials will be placed in sterile, leak-proof, and properly labeled containers. During laboratory processing, researchers will use appropriate PPEs such as laboratory gown, disposable nitrile gloves, hair cap, N95 face mask, shoe cover, and face shield. The researchers, prior to experimentation proper, will practice

appropriate disinfection procedures in work surfaces including the use of 1:10 dilution of 5.25% sodium hypochlorite with water, which both require at least 10-15 minutes contact time, Lysol spray, and the use of Class II biosafety cabinets (BSL 2) when handling potentially contaminated materials.

In case of accidental spillage or exposure, the researchers are to perform the SPILL drill which includes, immediate stop, presuming contamination, informing others about the spill, localization of spilled material, and labeling area as contaminated. Spills will be covered with paper towels, treated with 0.5% sodium hypochlorite solution or 1:10 bleach dilution, applied from the outside in the direction of the center, and let sit for 10-30 minutes. Following the required contact period, any residues will be thoroughly disinfected with 70% ethanol. All aforementioned procedures are strictly observed with use of personal protective equipment and in compliance with biosafety standards all throughout the research experimentation process.

Following the completion of the microbial analysis, all biological waste from the experimental procedures will be strictly disposed of in appropriate bins in compliance to biosafety guidelines, and will be labeled as biohazardous. All bacterial suspension samples, culture media, and disposable laboratory materials, will be autoclaved at 121°C and 15 psi for a minimum of 20 minutes to prevent environmental contamination and ensure laboratory safety prior to disposal to yellow biohazard bags (Lixuecai, 2025). Liquid wastes must be treated with 1% sodium hypochlorite solution with minimum 30 minutes contact time, while sharps

must be disposed of in puncture-proof containers and will also be autoclaved before final disposal. Final disposal will be managed through a licensed biohazardous waste service provider to prevent live microbial agents being released into the city waste stream.

Lastly, to ensure the safety of the researchers, a protocol will be followed to immediately suspend laboratory activities, secure all cultures and notify the TUA-IBC (Institutional Biosafety Committee) in the event of detection or suspected presence of high-risk pathogens beyond Biosafety Level 2.

Reference

Afonso. A. C., Saavedra, M. J., Gomes, I. B., Simões, M., & Simões, L. C. (2025). Current microbiological challenges in drinking water. Journal of Water Process Engineering, 72, 107614. https://doi.org/10.1016/j.jwpe.2025.107614

Alfonso, S. M., Kazama, S., & Takizawa, S. (2022). Inequalities in access to and consumption of safely managed water due to socio-economic factors: Evidence from Quezon City, Philippines. Current Research in Environmental Sustainability, 4, 100117. https://doi.org/10.1016/j.crsust.2021.100117

Angelakis, A. N., Voudouris, K., & Tchobanoglous, G. (2025). Water supply systems: Past, present challenges, and future sustainability prospects. Land, 14(3), 619. https://doi.org/10.3390/land14030619

Anokhe, A., & Kalia, V. (2022). Fermentation pathway by Methyl Red and Voges Proskauer (MR/VP) test. ResearchGate. https://www.researchgate.net/publication/356128142_Fermentation_Pathway_by_Methyl_Red_and_Voges_Proskauer_MRVP_Test

Arinao Murei, Ilunga Kamika, & Maggy Ndombo Benteke Momba. (2024). Selection of a diagnostic tool for microbial water quality monitoring and management of faecal contamination of water sources in rural communities. Science of the Total Environment, 906, 167484–167484. https://doi.org/10.1016/j.scitotenv.2023.167484

Banar, M., Kamyab, H., Torkashvand, N., Salehi, T. Z., Sepehrizadeh, Z., Shahverdi, A. R., Pourmand, M. R., & Yazdi, M. H. (2024). Protocols for Isolation, Characterization, and Therapeutic Application of Bacteriophages for MRSA Mastitis. Protocols.io. https://www.protocols.io/view/protocols-for-isolation-characterization-and-thera-dqat5sen.html

Barmase, M. (2025). Development of a Multi-Layer Herbal Water Filter for Bacterial Removal - A Review. *International Journal for Research in Applied Science and Engineering Technology*, *13*(5), 6432–6440. https://doi.org/10.22214/ijraset.2025.71836

Bekoe, E.M.O., Hayford, D., Quarcoo, G. et al. Groundwater microbial quality and risk evaluation in a rapidly transforming urbanized area of Northern Ghana. BMC Environ Sci 2, 9 (2025). https://doi.org/10.1186/s44329-025-00022-w

Bhatti, S., Gadhi, T. A., Mahar, R. B., Ali, I., Channa, N., Kori, J. A., & Bonelli, B. (2023). The Identification of Selective Pathogenic Microbial Community Biofilms in Different Distribution Pipeline Materials and Their Disinfection Kinetics. Water, 15(23), 4099. https://doi.org/10.3390/w15234099

Charles, K. J., Nowicki, S., Ong, L. A., Johnston, D., & Mahmud, Z. H. (2025). *Interpreting drinking water quality samples: Understanding contamination pathways at the point of collection*. **PLOS Water**, 4(10), e0000373. https://doi.org/10.1371/journal.pwat.0000373

Chavan, D. D., Khatoon, H., Anokhe, A., & Kalia, V. (2022, January 5). *Oxidase test: A biochemical method in bacterial identification*. ResearchGate; unknown. https://www.researchgate.net/publication/357604270_Oxidase_test_A_biochemical_method_in_bacterial_identification

Choudhary, D., Jawa, C., Dhuria, D., Sain, M., Mehara, M., & Kalwaniya, M. K. (2022). Prevalence of Staphylococcus aureus in raw fish meat samples and assessment of the coagulase test in the identification of Staphylococcus aureus isolates. The Pharma Innovation Journal, 11(7), 1324–1327. https://www.thepharmajournal.com

Corpuz, J. C. G. (2025). Wastewater management in the Philippines: A call for localized solutions and policy integration. Environmental Health Insights, 19, 11786302241311529. https://doi.org/10.1177/11786302241311529

Danert, K. (2022). Handpump standards, quality and supply chains with Zambia case study. Action research on handpump component quality and corrosion in sub-Saharan Africa'. *Stop the Rot Report III*. https://doi.org/10.13140/rg.2.2.18986.70089

Department of Health. (2022). Philippine national standards for drinking water (PNSDW).https://doh.gov.ph/wp-content/uploads/2023/08/Administrative-Order-No.-2017-0010-Philippine-National-Standards-for-Drinking-Water.pdf

Dockery, G. D. (2012). Aseptic techniques. In *Elsevier eBooks* (pp. 53–68). https://doi.org/10.1016/b978-0-7020-3136-6.00007-2

Dong, Y., Jiang, Z., Hu, Y., Jiang, Y., Tong, L., Yu, Y., Cheng, J., He, Y., Shi, J., & Wang, Y. (2024). Pathogen contamination of groundwater systems and health risks. Critical Reviews in Environmental Science and Technology, 54, 267–289. https://doi.org/10.1080/10643389.2023.2236486

Erdei-Tombor, P., Kiskó, G., & Taczman-Brückner, A. (2024). Biofilm Formation in Water Distribution Systems. Processes, 12(2), 280. https://doi.org/10.3390/pr12020280

Francisco, G. S., Daniel, E., Ayen, J., Angela, C., & Stefanie, M. (2024). Potability Assessment of the Deep Well Water (Tubig Poso) in the Ayta Community of Barangay Putingkahoy, Rosario, Batangas. Animo Repository. https://animorepository.dlsu.edu.ph/conf_shsrescon/2024/paper_see/2/

Hafezi, A., & Khamar, Z. (2024). The Method and Analysis of Some Biochemical Tests Commonly Used for Microbial Identification: A Review. Comprehensive Health and Biomedical Studies, 3(2). https://doi.org/10.5812/chbs-160199

Impact of biofilm in distribution water systems on water characteristics and attempts for treatment. (2025). https://rdcu.be/e2O3C

Infanta (Quezon) Water District. (2024). Operations Manual – Rev 3 2024. https://iqwd.gov.ph/wp-content/uploads/2024/04/Operations-Manual-Rev3-2024.pdf

Işıl, Ç., Koydemir, H. C., Eryilmaz, M., de Haan, K., Pillar, N., Mentesoglu, K., Unal, A. F., Rivenson, Y., Chandrasekaran, S., Garner, O. B., & Ozcan, A. (2025). Virtual Gram staining of label-free bacteria using dark-field microscopy and deep learning. Science Advances, 11(2). https://doi.org/10.1126/sciadv.ads2757

Ismail, H. M., Muse, A. H., Hassan, M. A., Muse, Y. H., & Nadarajah, S. (2024). Analyzing Unimproved Drinking Water Sources and Their Determinants Using Supervised Machine Learning: Evidence from the Somaliland Demographic Health Survey 2020. *Water*, *16*(20), 2986. https://doi.org/10.3390/w16202986

Ito, H., Tomura, Y., Oshida, J., Fukui, S., Kodama, T., & Kobayashi, D. (2023). The role of gram stain in reducing broad-spectrum antibiotic use: A systematic literature review and meta-analysis. *Infectious Diseases Now*, *53*(6), 104764. https://doi.org/10.1016/j.idnow.2023.104764

Kelly, E., Cronk, R., Fisher, M., & Bartram, J. (2021). Sanitary inspection, microbial water quality analysis, and water safety in handpumps in rural sub-Saharan Africa. *Npj Clean Water*, *4*(1). https://doi.org/10.1038/s41545-020-00093-z

Lestari, D., Haliza, N., & Setiawan, A. (2024). Effect of pH variations in eosin methylene blue agar (EMBA) medium on E. coli growth. ResearchGate. https://www.researchgate.net/publication/380589238_Effect_of_pH_Variati

ons_in_Eosin_Methylene_Blue_Agar_EMBA_Medium_on_E_coli_Growth

Li, H., McManus, C., & Cronk, R. (2025). Factors influencing water point functionality in Liberia: A regression and Bayesian network analysis. Sustainability, 17(19), 8928. https://doi.org/10.3390/su17198928

Lixuecai. (2025, November 10). *Why 121°C is the Standard Temperature for Autoclaving: The Science Behind Effective Sterilization*. Sada Medical. https://www.sadamedical.com/blogs/news/why-121-c-is-the-standard-temperature-for-autoclaving-the-science-behind-effective-sterilization?srsltid=AfmBOorR3K645ou01dtBlH0AAUjRKDda-8sEaJ1xwVLo3skp8D5lihZ8

Luft, J. A., Jeong, S., Idsardi, R., & Gardner, G. (2022). Literature reviews, Theoretical frameworks, and Conceptual frameworks: An Introduction for New Biology Education Researchers. CBE—Life Sciences Education, 21(3). https://doi.org/10.1187/cbe.21-05-0134

Madamanchi, Divya. Water, sanitation, and hygiene (WASH) strategies in the fight against waterborne diseases in the young and old. Journal of Family Medicine and Primary Care 13(8):p 3481-3482, August 2024. | DOI: 10.4103/jfmpc.jfmpc_718_24

Molina, V. B., Sison, O. T., Medina, J. R. C., Lumangaya, C. R., Ayes, C. N. P., Joe, J. A. S., & Belizario, V. Y., Jr. (2021). *Water, sanitation and hygiene practices in the Philippines: Meeting national and global targets at the*

local level. Journal of Environmental Science and Management, 24(1), 1–14

Kanyagui, M., Sharma, J., & Mishra, N. (2023). Assessing the implications of water provisioning and its quality on health outcomes: A micro level study in India. Research Square. https://doi.org/10.21203/rs.3.rs-3119391/v1

Knecht, A. E., Ettenauer, J., Posnicek, T., Taschl, M., Helmecke, M., Haller, H., Gölß, S., & Brandl, M. (2023). Differences in Microbial Communities in Drinking Water from Conventional Electronic and Manual Taps in Dependence on Stagnation and Flushing Cycles. Water, 15(4), 784. https://doi.org/10.3390/w15040784

MacAllister, D. J., Nedaw, D., Kebede, S., Mkandawire, T., Makuluni, P., Shaba, C., Okullo, J., Owor, M., Carter, R., Chilton, J., Casey, V., Fallas, H., & MacDonald, A. M. (2022). Contribution of physical factors to handpump borehole functionality in Africa. Science of the Total Environment, 851, 158343. https://doi.org/10.1016/j.scitotenv.2022.158343

Mbachu, A., Ani, C., & Mbachu, N. (2022). ASSESSMENT OF BACTERIOLOGICAL AND PHYSICOCHEMICAL STATUS OF WELL WATERS IN EMENE ENUGU, NIGERIA. *Bacterial Empire*, *5*(2), e412. https://doi.org/10.36547/be.412

Mimura, M., Koizumi, Y., Wada, M., Ichijo, T., Uchii, K., & Nasu, M. (2023). Microbial Water Quality Assessment of Private Wells Using 16S rRNA Gene Amplicon Sequencing with a Nanopore Sequencer. Biological &

pharmaceutical bulletin, 46(2), 263–271. https://doi.org/10.1248/bpb.b22-00690

Niboucha, N., Goetz, C., Sanschagrin, L., Fontenille, J., Fliss, I., Labrie, S., & Jean, J. (2022). Comparative study of different sampling methods of biofilm formed on stainless-steel surfaces in a CDC biofilm reactor. *Frontiers in Microbiology, 13,* 892181. https://doi.org/10.3389/fmicb.2022.892181

Nguyen, T. A., Le, D. C., Nguyen, T. N., Schmalz, B., & Tran, L. L. (2023). Influences of key factors on river water quality in urban and rural areas: A review. Case Studies in Chemical and Environmental Engineering, 8, 100424. https://doi.org/10.1016/j.cscee.2023.100424

Odeyemi, A. T., Akinjogunla, O. J., Ajayi, A. O., Thomas, R. M., & Adeoye, O. O. (2023). Stream and well water samples from two rural communities in Ekiti State, Nigeria: assessment of physicochemical parameters, bacteriological quality and public health significance. *Acta Scientiarum Biological Sciences*, *45*, e64296. https://doi.org/10.4025/actascibiolsci.v45i1.64296

Oliveira, I. M., Gomes, I. B., Simões, L. C., & Simões, M. (2024). A review of research advances on disinfection strategies for biofilm control in drinking water distribution systems. Water research, 253, 121273. https://doi.org/10.1016/j.watres.2024.121273

Panganiban, M. J. T., Rosario, I. R. A., Sabino, N. G., (2024). Multidrug-resistant Escherichia coli from Artesian Well Waters in Los Baños, Laguna, the Philippines. In Philippine Journal of Science (Vols. 153–2, pp. 619–629)

[Journal-article]. https://philjournalsci.dost.gov.ph/wp-content/uploads/2024/04/multidrug-resistant-E-coli-from-artesian-well-waters-in-Los-Banos-Laguna_.pdf

Papciak, D., Domoń, A., Zdeb, M., Tchórzewska-Cieślak, B., Konkol, J., & Sočo, E. (2022). Mechanism of biofilm formation on installation materials and its impact on the quality of tap water. *Water, 14*(15), 2401. https://doi.org/10.3390/w14152401

Paray, A.A., Singh, M.R., & Mir M. (2023). Gram Staining: a Brief Review. International Journal of Research and Review, 10(9), 336–341. https://doi.org/10.52403/ijrr.20230934

PathWest Laboratory Medicine WA. (2021). Collection and Transportation Instructions for Water Sampling: MWP119 Version 2.0. https://www.pathwest.health.wa.gov.au/~/media/PathWest/Documents/Our-Services/Commercial-Services/Food-and-Water-Testing/MWP119-Guide-to-Collecting-and-Transporting-Water-Samples.pdf

Philippine Institute for Development Studies. (2024). Persistent diarrhea cases in PH highlight gaps in clean water access, PIDS study reveals [Press release]. https://www.pids.gov.ph/details/news/press-releases/persistent-diarrhea-cases-in-ph-highlights-gaps-in-clean-water-access-pids-study-reveals

Rao, P., & Mulky, L. (2023). An overview of microbiologically influenced corrosion on stainless steel. *ChemBioEng Reviews, 10*(5), 829–840. https://doi.org/10.1002/cben.202300009

Research SOP. (2023, May 30). *Standard operating procedure for surface swab test*. Research SOP. https://www.researchsop.com/2023/05/standard-operating-procedure-for_30.html

Rivett, M. O., Tremblay-Levesque, L.-C., Carter, R., Thetard, R. C. H., Tengatenga, M., Phoya, A., Mbalame, E., Mchilikizo, E., Kumwenda, S., Mleta, P., Addison, M. J., & Kalin, R. M. (2022). Acute health risks to community hand-pumped groundwater supplies following Cyclone Idai flooding. Science of The Total Environment, 806(Pt. 2), 150598. https://doi.org/10.1016/j.scitotenv.2021.150598

Rural Water Supply Network. (2024). Handpump technology. Rural Water Supply Network. https://rwsn.blog/category/sustainable-groundwater-development/handpump-technology/

Ryu, H., Addor, Y., Brinkman, N. E., Ware, M. W., Boczek, L., Hoelle-Schwalbach, J., Mistry, J. H., Keely, S. P., & Villegas, E. N. (2021). Understanding microbial loads in wastewater treatment works as source water for water reuse. Water, 13(11), 1452. https://doi.org/10.3390/w13111452

Salawu, R. O., Aina, S., Bolatito, S., & Masibo, S. (2023). Theoretical and conceptual frameworks in research: Conceptual clarification. European

Chemical Bulletin, 12(12), 2103–2117. https://doi.org/10.48047/ecb/2023.12.12.139

Sanchez, P. A., Sobremisana, M., de Jesus, A. L. M., Sobremisana, A., Mozo, M. J., Gigantone, C., Gapan, E. Z. S., Lomod, J. A., Navarro, D. K., & Lozada, R. (2024). Vulnerability of freshwater resources in the Philippines and its regions. Journal of Environmental Science and Management, 27(1), 61–79. https://doi.org/10.47125/jesam/2024_1/06

Senate Economic Planning Office. (2023, August). The state of water: At a glance (AAG 23-03). Senate of the Philippines. https://legacy.senate.gov.ph/publications/SEPO/State%20of%20Water%20AAG_August%202023.pdf

Singh, N. P., Kulshrestha, M., Shukla, N., Mishra, V., & Kulshreshtha, M. (2021). Management of risk factors for breaking localised pathways of microbial contamination in tubewells with handpump: a case study from India. Modeling Earth Systems and Environment, 8(2), 2513–2523. https://doi.org/10.1007/s40808-021-01242-0

Tareq, M. A., Rifat, M. T., Islam, M. S., Mondol, P., & Khan, M. S. I. (2024). Microbial quality of ready-to-eat food contact surfaces in local restaurants of Patuakhali district, Bangladesh. Asian Journal of Medical and Biological Research, 10(1), 71–79. https://doi.org/10.3329/ajmbr.v10i1.71232

Thaniem, M., Prakash, A., & Muniyandi, M. (2022). The Impacts on Consumer Health from Dug Well Contamination - A Crucial Drinking Water Source.

International Journal for Research in Applied Science and Engineering Technology, *10*(8), 1318–1327. https://doi.org/10.22214/ijraset.2022.46414

Tripathi, N., Sapra, A., & Zubair, M. (2025). Gram Staining. National Library of Medicine; StatPearls Publishing. https://www.ncbi.nlm.nih.gov/books/NBK562156/

Tuddao, V. B., Jr., & Gonzales, E. (n.d.). Updates of water environment governance in the Philippines. In *The 19th Annual Meeting in Hayama, Japan*. https://wepa-db.net/wp-content/uploads/2024/02/PM07_0130_country-update_PHL-1.pdf

Ulep, V. G., Talamayan, J. T., Casas, L. D., Villaseñor, J. M., & Bacatan, E. (2024). The last-mile challenge: Water, sanitation, and hygiene (WASH) in the Philippines (PIDS Discussion Paper Series No. 2024-19). Philippine Institute for Development Studies. https://doi.org/10.62986/dp2024.19

Unson, J. (2024, January 3). Tubig sa mga poso nakontamina 3 tepok, 21 pa naospital sa diarrhea outbreak. Philstar.com; Pilipino Star Ngayon. https://www.philstar.com/pilipino-star-ngayon/probinsiya/2024/01/04/2323306/tubig-sa-mga-poso-nakontamina-3-tepok-21-pa-naospital-sa-diarrhea-outbreak

Velasco, J. (2023, March 24). *11M PAMILYA, POSO PINAGKUKUNAN NG TUBIG*. https://saksipinas.com/kalikasan/11-m-pamilya-poso-lang-pinagkukunan-ng-tubig/

World Health Organization. (2023, September 13). Drinking-water. World Health Organization. https://www.who.int/news-room/fact-sheets/detail/drinking-water

World Health Organization. (2023). Drinking-water [Fact sheet]. https://www.who.int/news-room/fact-sheets/detail/drinking-water

World Bank (2024). Preventing Rapid Corrosion of Handpumps Through Regulations. World Bank. http://documents.worldbank.org/curated/en/099052424151038179

World Health Organization. (2024). Sanitation. World Health Organization. https://www.who.int/news-room/fact-sheets/detail/sanitation

Yamaji, M., Kusama, Y., Kamimura, K., & Maihara, T. (2021). Difficulty of morphological diagnosis in Gram staining. BMJ Case Reports, 14(8), e245349. https://doi.org/10.1136/bcr-2021-245349

Zhu, Z., Shan, L., Hu, F., Li, Z., Zhong, D., Yuan, Y., & Zhang, J. (2020). Biofilm formation potential and chlorine resistance of typical bacteria isolated from drinking water distribution systems. RSC Advances, 10(52), 31295–31304. https://doi.org/10.1039/d0ra04985a